

Equipment Datasheet — Setpoints & Limits

Distillation Column D-201 / Reboiler E-201 / Condenser C-201

Sheet DS-D201-02, Rev 2, 2026-02-20. Design basis for control and safe operating limits.

Design Data

Item	Value	Item	Value
No. of theoretical stages	18	Feed stage	Stage 10 (from top)
Design pressure	2.0 bar a	Design temperature	130 °C
Column diameter	1.2 m	Reboiler type	Thermosiphon, LP steam
Feed composition (design)	10% v/v EtOH	Distillate spec	95% v/v EtOH

Control Setpoints (normal)

Loop	Controlled variable	Setpoint	Notes
PIC-201	Top pressure	1.15 bar a	Vent/pressure control
FIC-202	Reflux flow (ratio)	L/D = 3.0	Ratio to distillate
TIC-204	Reboiler steam	~2.4 t/h	Trims bottoms temp
FIC-200	Feed flow	12.0 m³/h	Beer feed

Safe Operating Limits (do not exceed without approved deviation)

Variable	Min	Max	Trip
Top pressure (bar a)	1.05	1.30	1.50 (steam cut)
Reflux ratio L/D	2.5	3.8	–
Reboiler steam (t/h)	1.8	3.0	3.3 (limit)
Bottoms temp (°C)	98	107	112 (steam cut)
Distillate purity (% v/v)	94.0	–	93.5 divert to slops